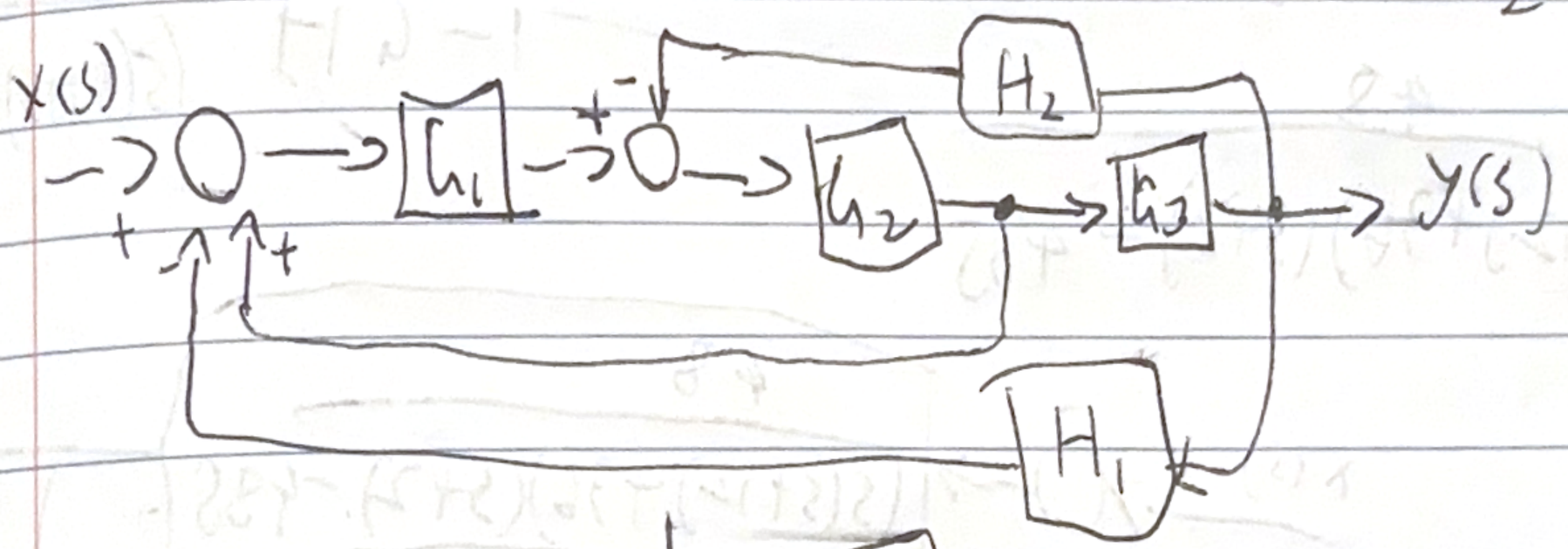
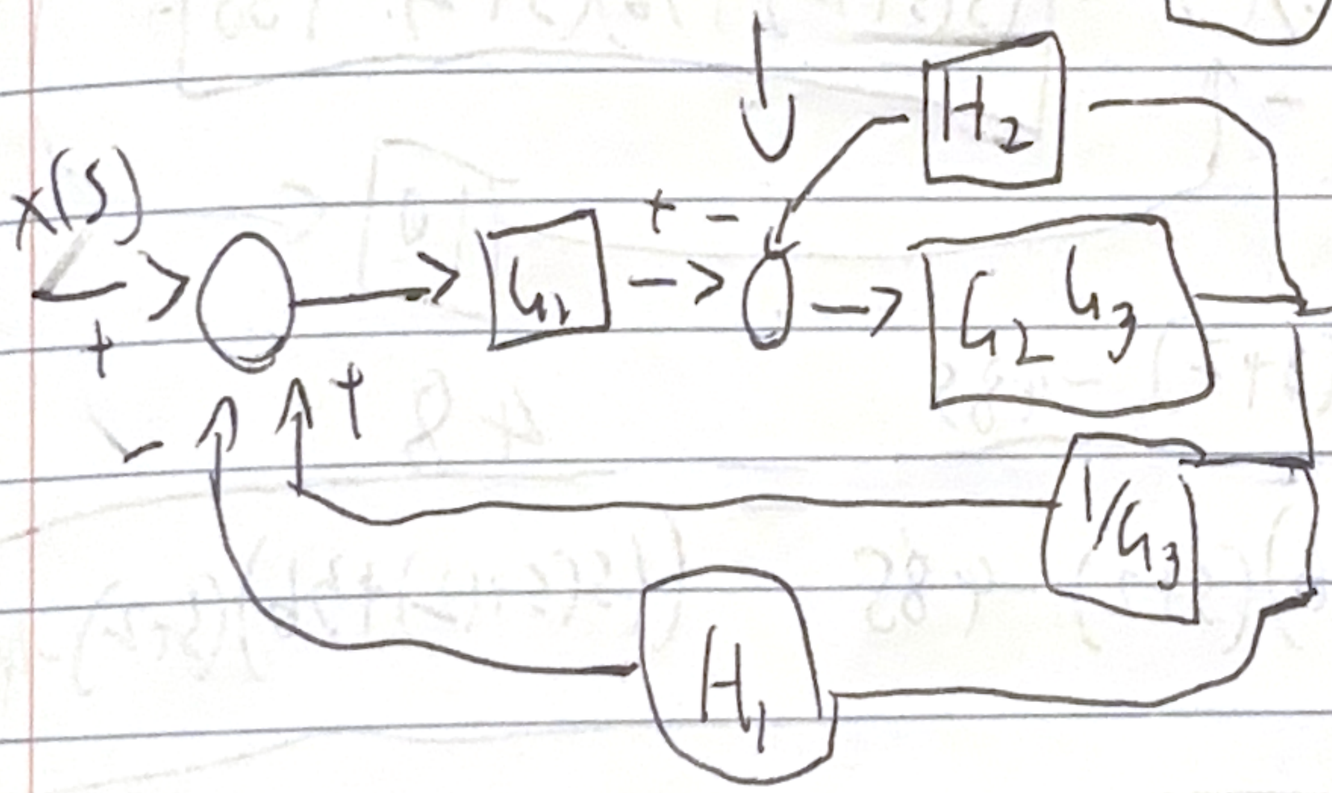
(8) $G_1(s) = \frac{4}{s+2}$, $G_2(s) = \frac{12}{s+12}$, $G_3(s) = \frac{1}{s}$, $H_1(s) = 10$, $H_2(s) = 3$



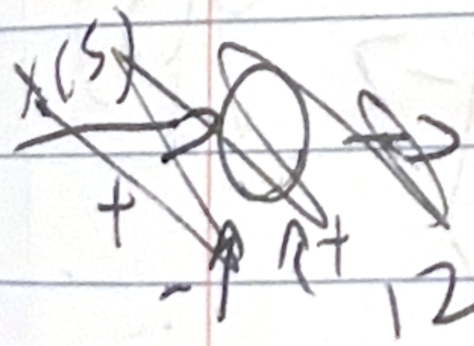

$G = G_2 G_3$

$H = H_2$

$$\frac{G_2 G_3}{1 + G_2 G_3 H_2} = \frac{\left(\frac{12}{s(s+12)}\right)}{1 + \left(\frac{12}{s(s+12)}\right)3}, \quad \frac{s(s+12)}{s(s+12)}$$

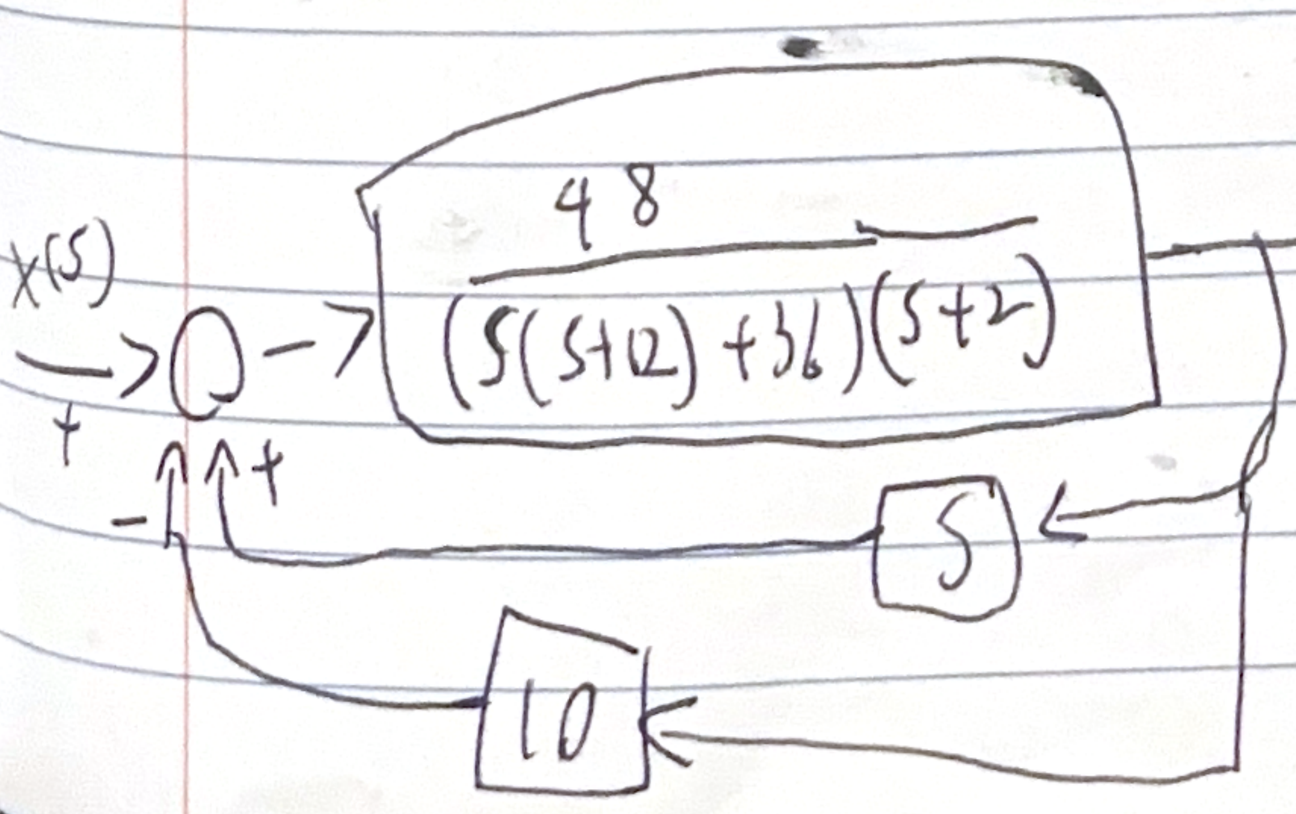
$$= \frac{12}{s(s+12) + 36}$$

multiply by $G_1$




$$\frac{12}{s(s+12)+36} \cdot \frac{4}{s+2}$$

$$= \frac{48}{(s(s+12)+36)(s+2)}$$




$$\frac{4}{s+2}$$

continue →

(8)

continued

$G = \frac{48}{(s(s+12)+36)(s+2)}$ $\quad H = 5$

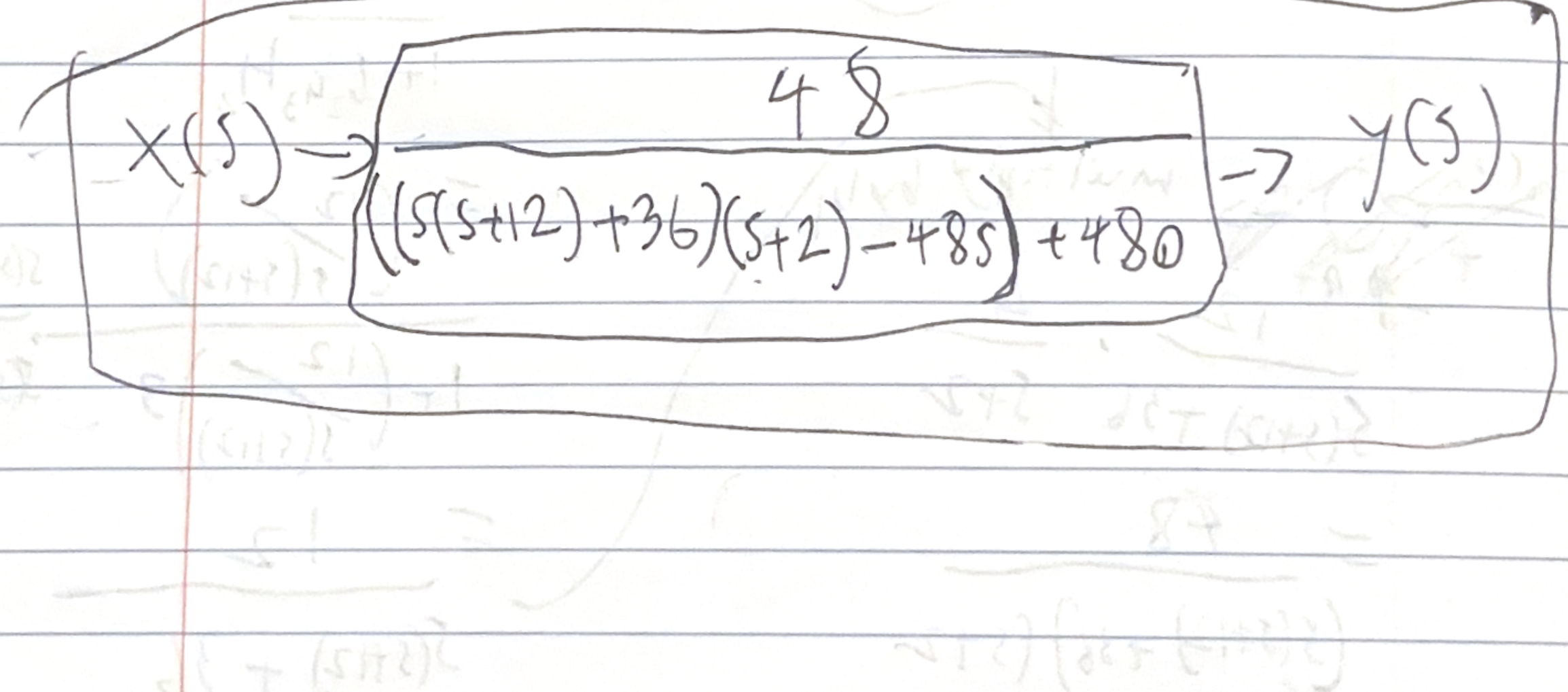
$$\frac{48}{(s(s+12)+36)(s+2)-485}$$

$$\frac{G}{1-GH} = \frac{\frac{48}{(s(s+12)+36)(s+2)}}{1-\frac{48}{(s(s+12)+36)(s+2)}\cdot 5} = \frac{\frac{48}{(s(s+12)+36)(s+2)}}{\frac{(s(s+12)+36)(s+2)-485}{(s(s+12)+36)(s+2)}}$$

$$= \frac{48}{(s(s+12)+36)(s+2)-485}$$

$G = ?$

$H = 10$




$$\frac{G}{1-GH} = \frac{\frac{(s(s+12)+36)(s+2)-485}{(s(s+12)+36)(s+2)-485}}{1-\frac{(s(s+12)+36)(s+2)-485}{(s(s+12)+36)(s+2)-485}\cdot 10} = \frac{48}{((s(s+12)+36)(s+2)-485)+480}$$